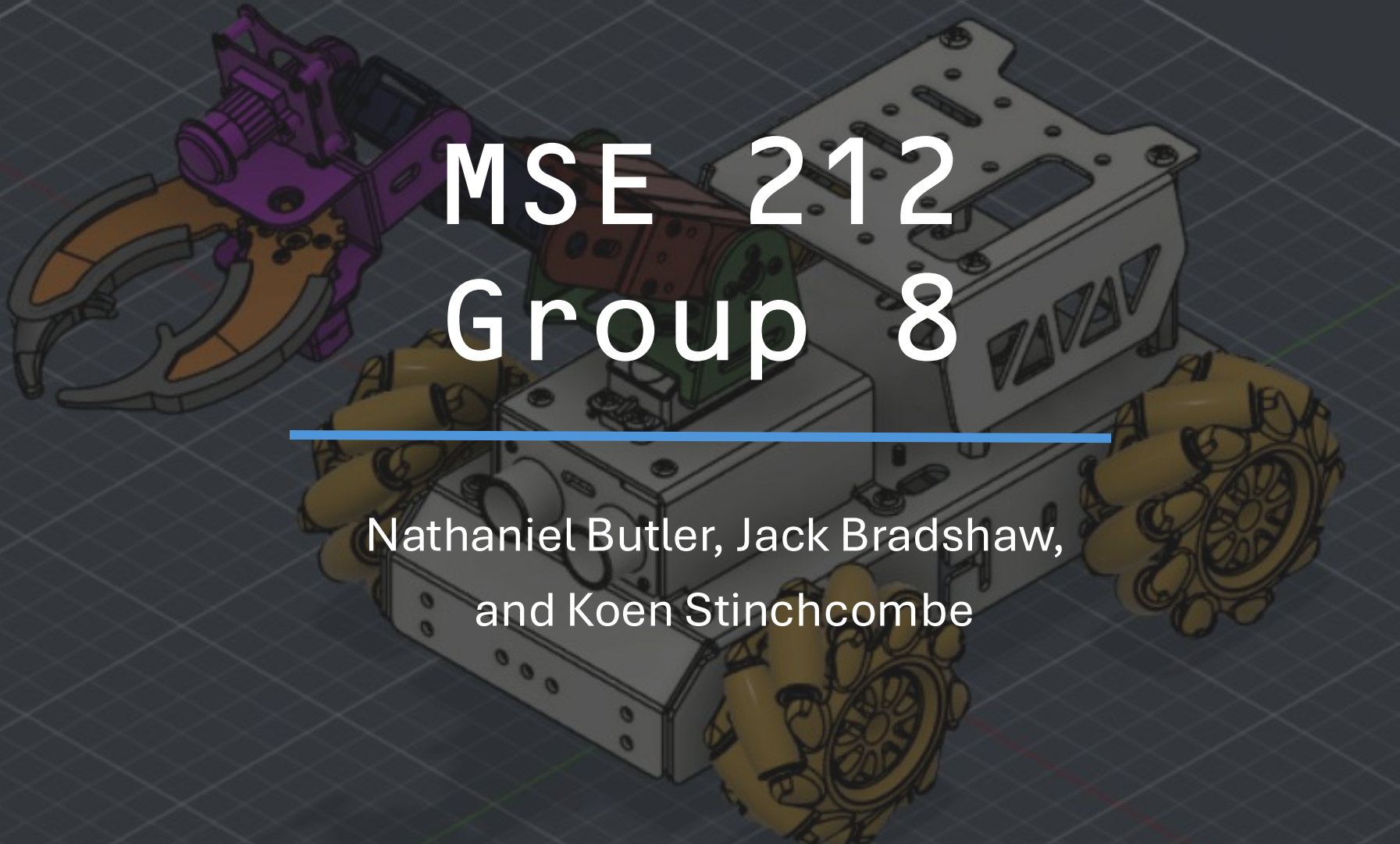


A 3D CAD model of a robotic gripper and base assembly. The gripper is purple and orange, mounted on a grey base. The base has four yellow wheels. The model is shown on a grey grid background.

MSE 212 Group 8

Nathaniel Butler, Jack Bradshaw,
and Koen Stinchcombe

Introduction

- Across four projects we designed, fabricated, and simulated a complete robot
- Starting from a step files we reverse engineered the parts of the robot
- Designed custom gripper, 3D printed it, and tested it physically
- Finally, we mirrored the robot's real arm movement in a digital twin simulation

Project 1 — SolidWorks reverse engineering

Import Hiwonder robot arm STEP files
Reconstruct 60–80 parts as SLDPRT → full SLDASM with GD&T drawings

geometry foundation

Project 2 — Chess piece design & gripper mechanism

CAD the target chess piece in SolidWorks
Design gripper mechanism geometry, joints, and attachment strategy

mechanism design → sleeve prototype

Project 3 — Gripper sleeve design & FDM prototyping

Design sleeve to mount on existing gripper, optimised for 3D printing
Print in PLA, test grip stability, iterate based on performance

validated hardware → digital twin

Project 4 — Digital twin (ROS + MQTT)

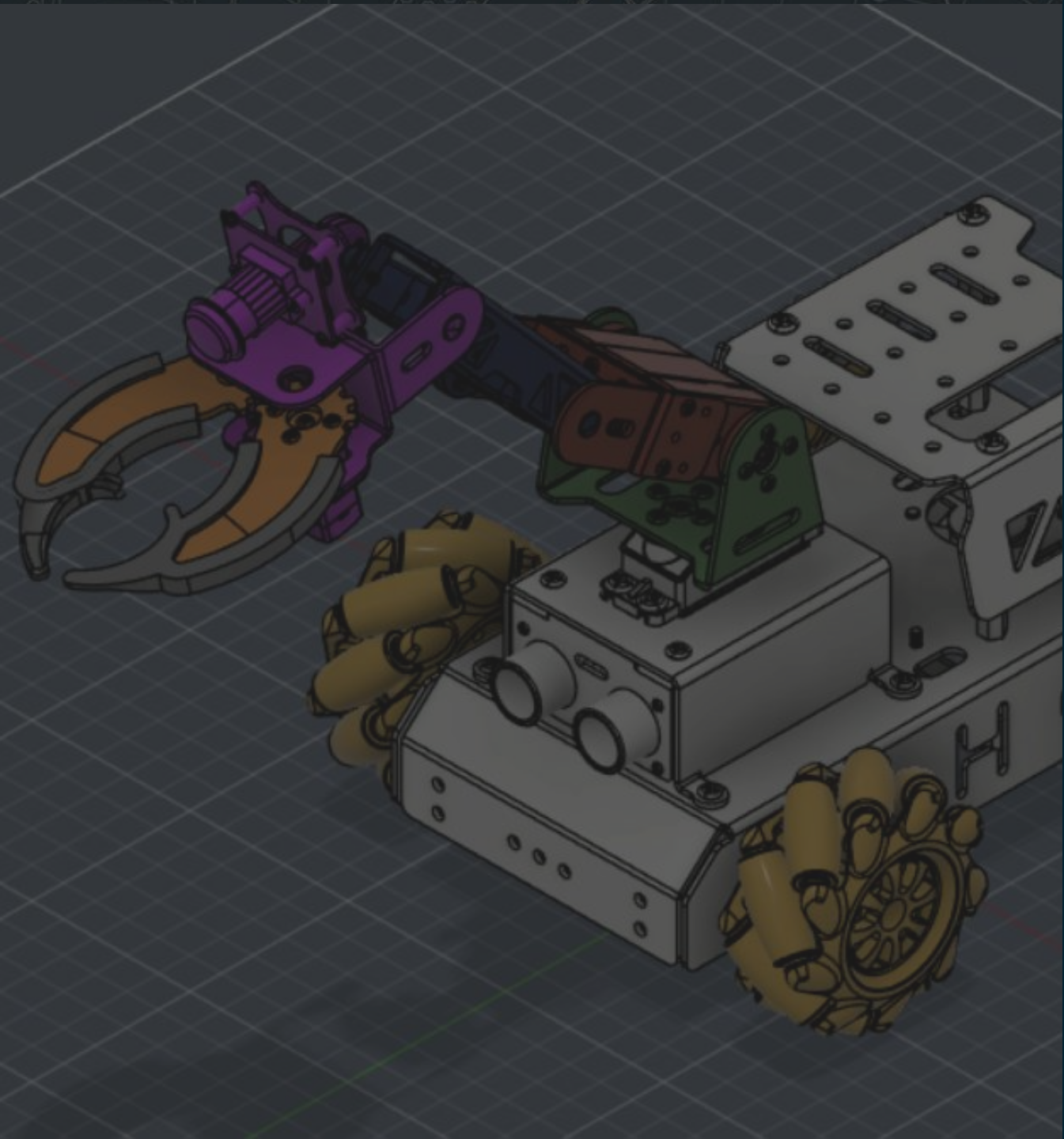
Export URDF/XACRO from Fusion 360, simulate in RViz2
MQTT bridges real arm (RPi publisher) to ROS simulation (subscriber)

Final Objective

- Pickup a chess piece with the HiWonder robot and run a digital sim of this process.

How?

- Breaking this objective into 4 projects
 1. Reverse-Engineering HiWonder
 2. 3d Scanning and Re-Eng Chess Piece
 3. Design and 3d print Custom Gripper
 4. Digitally Simulate HiWonder



Problems and Requirements

Project 1

Problem: STEP file are non-editable, need fully parametric model to work from

Requirements: Construct 60-80 parts accurately with drawing and a full assembly

Constraints: Use SolidWorks, geometry must match STEP file exactly

Project 3

Problem: Claw does not securely pick up chess piece, need gripper attachment

Requirements: Sleeve must mount onto claw, be 3D printable, hold chess piece without dropping

Constraints: FDM, PLA material, wall thickness

Project 2

Problem: No existing claw mechanism, need to CAD the claw and chess piece

Requirements: claw and chess piece must be accurately modeled in SolidWorks

Constraints: Claw geometry must match STEP file, Chess piece must be modeled from 3D scan

Project 4

Problem: Simulate arms motion with digital twin

Requirements: Mirror arm motion in real time with MQTT and ROS

Constraints: Software's (Fusion 360, ROS, MQTT, RViz2)

Background

- Goal: Reverse Engineer and improve upon the HiWonder Robot
- The labs for the first half of the course taught us how to utilize SolidWorks for use in the coming projects.
- For the scanning and modeling of the chess piece we had to learn:
 - the differences between meshes and step files
 - how to make successful scans using vanishing spray and multiple angles
- For the making of the custom gripper we had to learn:
 - Making manufacturable parts, for 3d printing key factors include tolerances, wall thickness being greater than the nozzle thickness, mindfulness of support removal/use

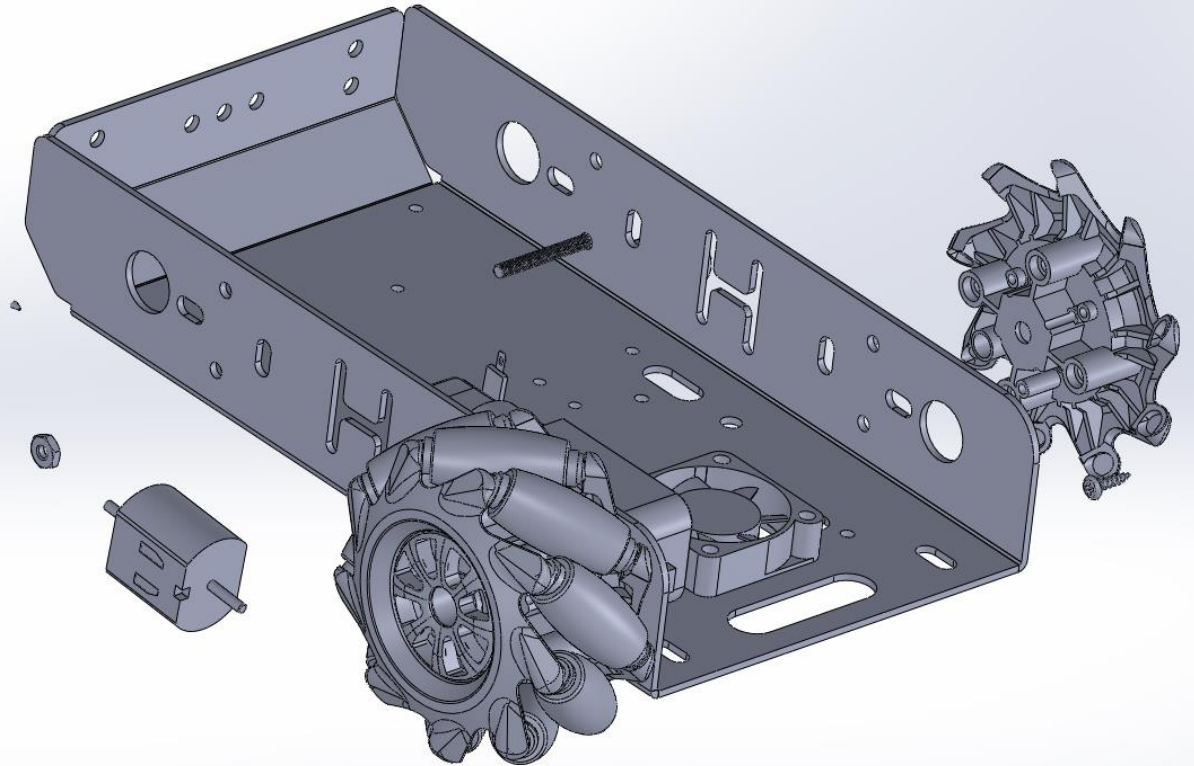
Background

- For the digital twin we had to learn the basics of how the ROS system, and MQTT protocol function
 - ROS: The Robot Operating System is a system that uses tools to accomplish everything needed to simulate a robot. Each tool has a different specialty while ROS oversees and manages them.
 - Rviz: visualizer to make the simulation viewable to users
 - Gazebo: physics simulator
 - Moveit: manages joints and robot fwd/rev kinematics
 - URDF: Unified Robot Description Format is a file that stores the linkage information of the robot.
 - MQTT: Message Queuing Telemetry Transport is a communication protocol used to send the real world status of the robot to the ROS system to allow simulation.

Project 1 Summary

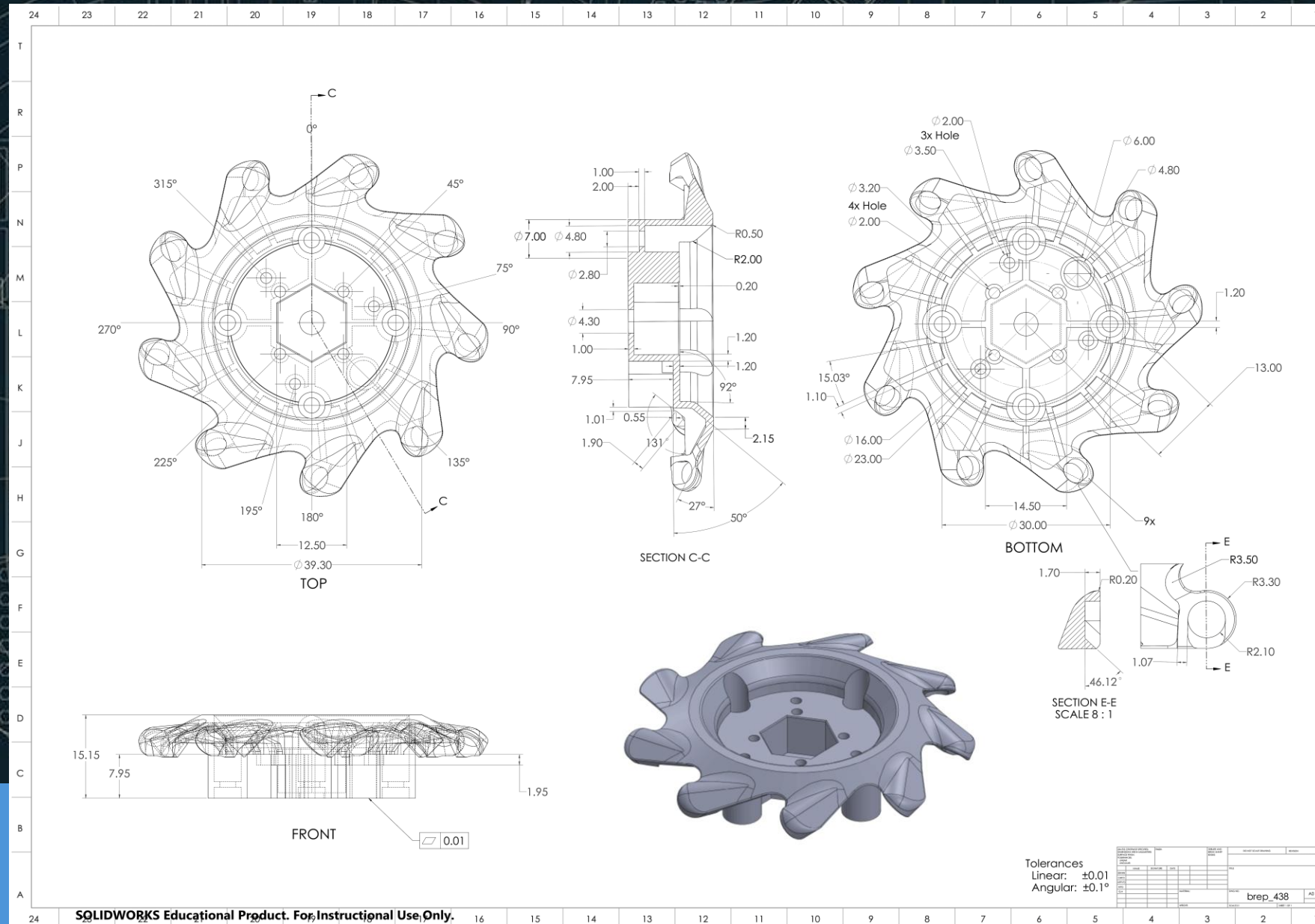
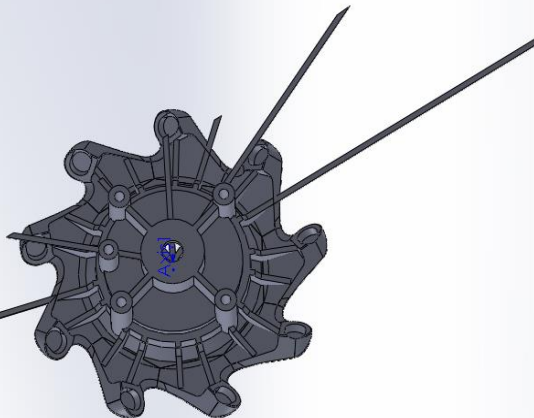
Method

- Took measurements and created sketches directly from STEP files
- Built initial CAD geometry based on extracted dimensions
- Fully constrain and dimension sketches



Project 1

- Rounding and tolerance errors caused intended symmetric geometry to behave inconsistently
- Created a single reference feature instead of multiple identical ones
- Used a circular body pattern (patterning solid bodies, not features)



Project 2 Summary

3D Scanning to CAD Design

- 3D Scanner used in the lab to collect point cloud information about the Chess Piece
- From this point cloud a mesh can be exported using the same software as the camera
- This Mesh is placed in solidworks where it is referenced and used to reverse engineer the initial piece.



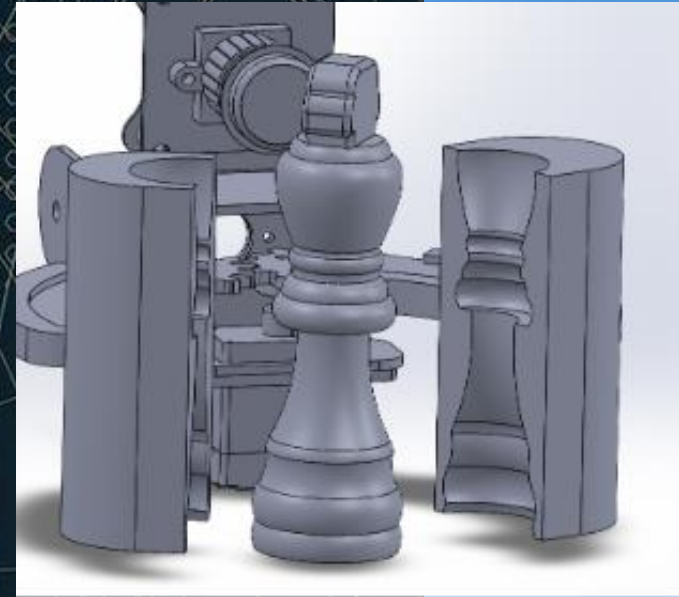
Point Cloud
(.xyz)

Mesh
(.stl)

Solidworks
(.sldprt)

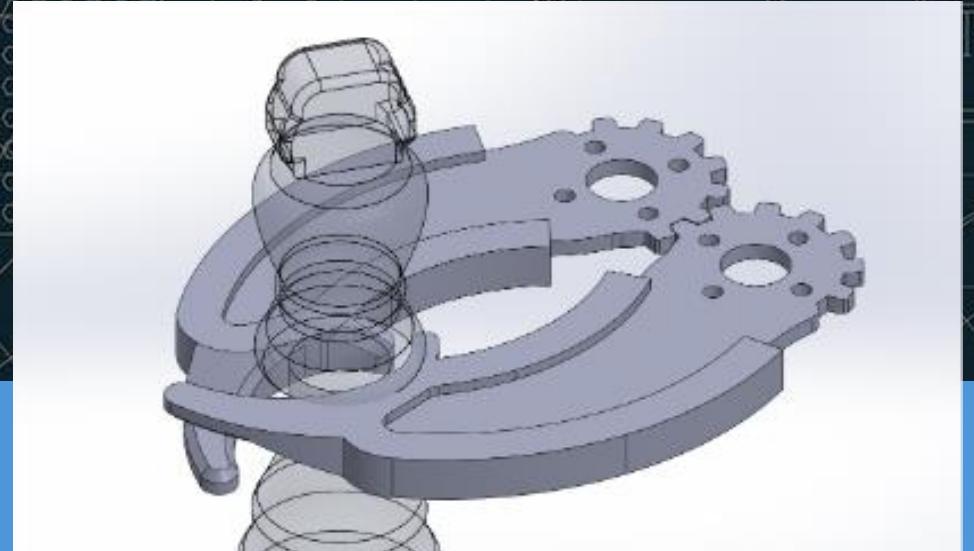
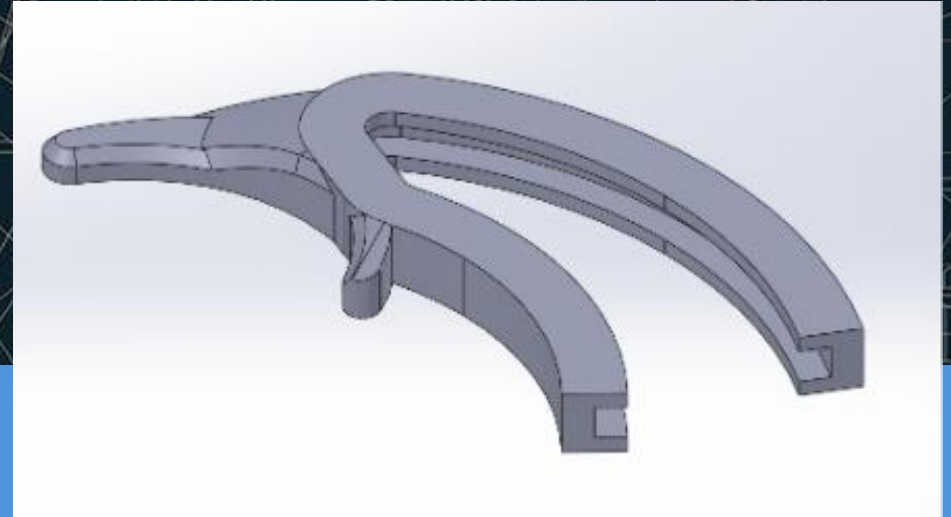
Project 3 Summary

- Mold-like fit matching most of the chess piece geometry
 - Ensures secure capture
 - Needs to be very accurate to ensure piece is picked up properly
-
- Length and width of square is slightly bigger than diameter of chess piece for margin of error
 - Overlap creates tight lock ensuring the chess piece does not fly out when moved



Project 3 Summary

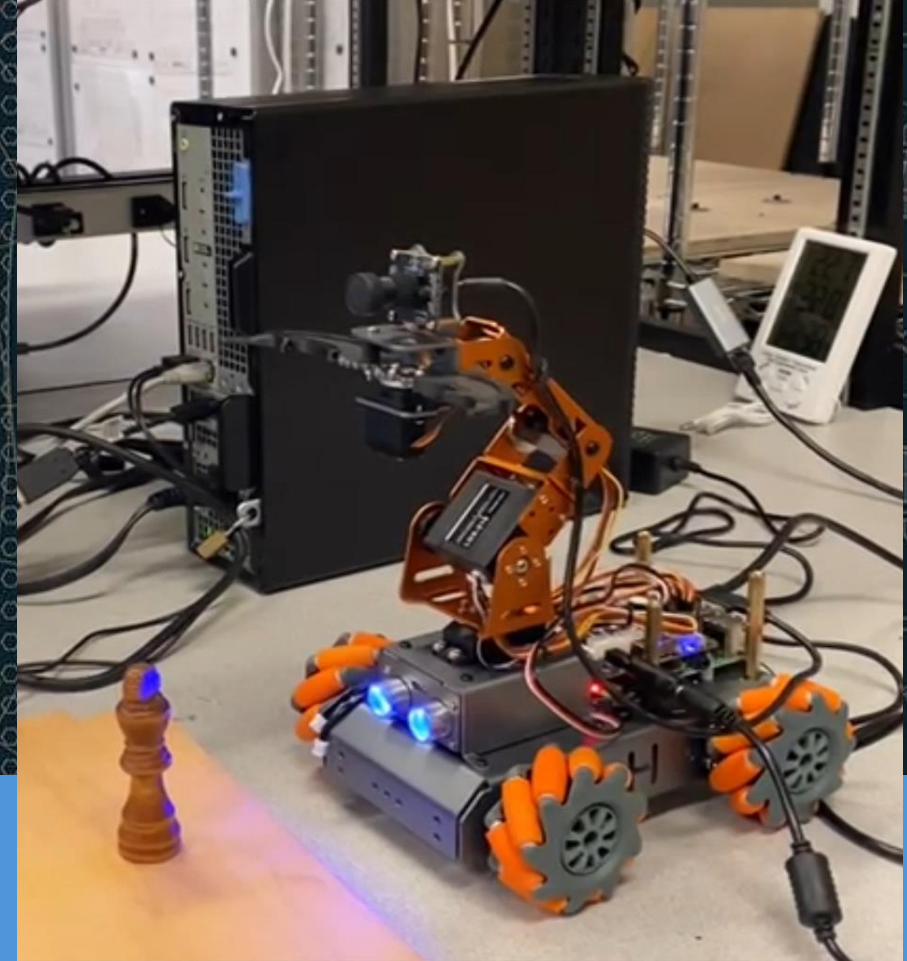
- Inner diameter matches the chess piece at the target pickup point
- Outer hook allows grabbing the piece from a distance and pulling it inward
- Overlapping hooks improve grip angle and pulling effectiveness
- Chamfered top helps guide the piece into optimal position
- Hook widens toward the base for increased strength and stiffness
- One flat side eliminates overhangs → no supports needed for 3D printing
- Symmetrical design (same part on both sides) reduces design, printing, and assembly complexity



Testing and Results

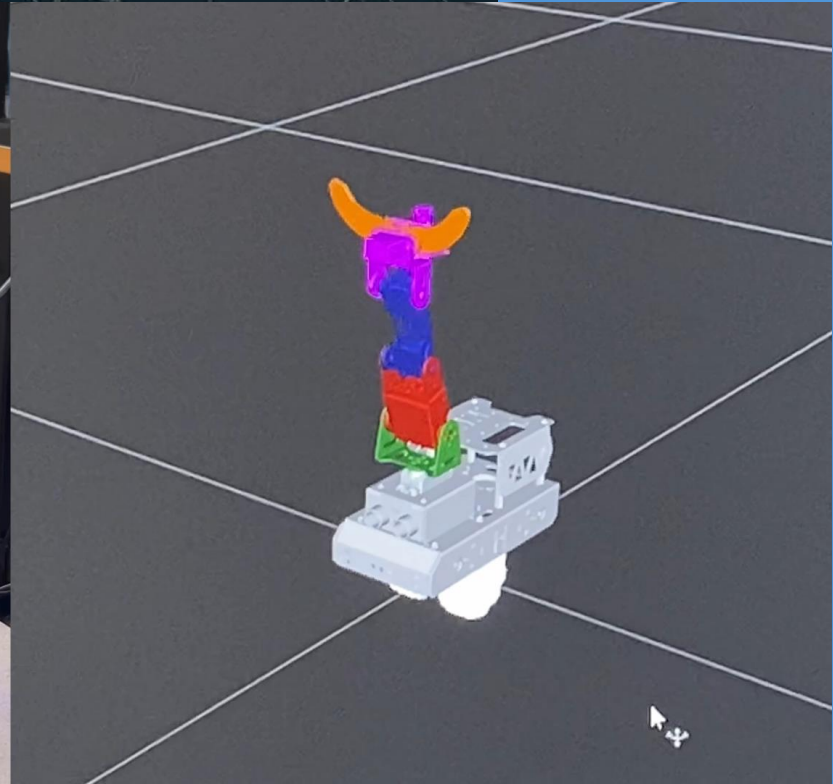
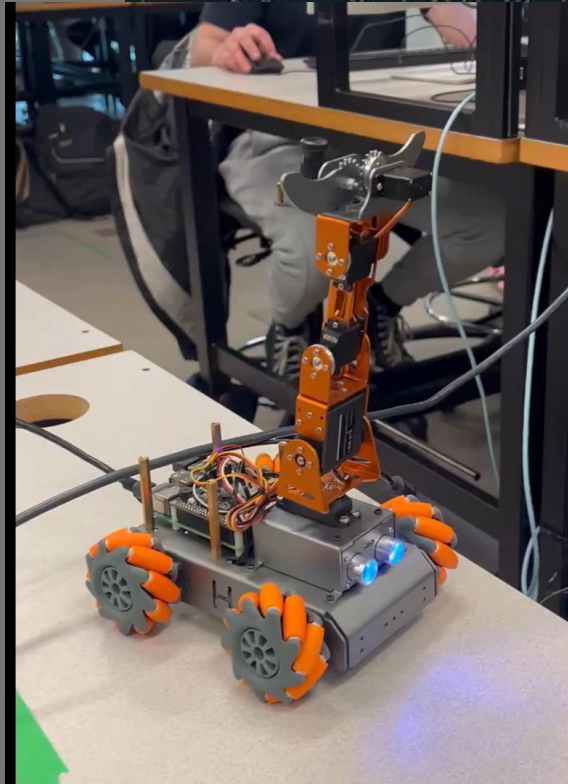
Test successful,

- Design chosen securely picks up piece and moves it around without slipping.
- Gripper designed held onto the HiWonder Robot loosely but still held on.
- Potential improvement being to tighten the tolerances so that there is a firmer attachment between the gripper and the robot.



Project 4 Progress

- Import robot into Fusion and configured links and joints
- Export URDF file from Fusion
- Loaded URDF file into RViz2 to create the live simulation of the physical robot
- robot_mqtt_bridge takes the MQTT data and feeds it to ROS, mirroring real arm movement
- Adjusted starting position of motors on RViz2 to line up simulation correctly
- Functional digital twin of the robot



Conclusion

Summary

- We successfully designed, fabricated, and simulated a robotic system capable of pick up a chess piece across 4 projects

Learning Outcomes

- We gained hands on experience with reverse engineering
- We applied engineering design process through CAD modeling, to prototyping, and testing
- We learned how ROS, MQTT, and URDF work together to create a digital twin of a physical robot

What Worked Well

- Gripper sleeve successfully mounted onto the claw and picked up the chess piece
- Digital twin successfully mirrored the physical arm's motion in RViz2 in real time

Future Improvements

- Refine gripper geometry for better performance (tighter fit to claw)
- Improve simulation accuracy and reduce latency between physical arm and the digital twin

The background is a dark blue-grey color with a complex, light-colored technical drawing or blueprint pattern. The pattern consists of numerous overlapping circles, arcs, and straight lines, some of which form rectangular shapes resembling architectural floor plans. The lines are thin and light grey or white, creating a dense, intricate web of geometric shapes across the entire frame.

Thank you

Any Questions?